Inflation

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The road towards 2% inflation

12-month PCE inflation in September 2026:

- Headline: 3.6%
- Ex Energy Goods: 3.0%
- Core: 3.1%

Annualized PCE inflation through the year in September 2026:

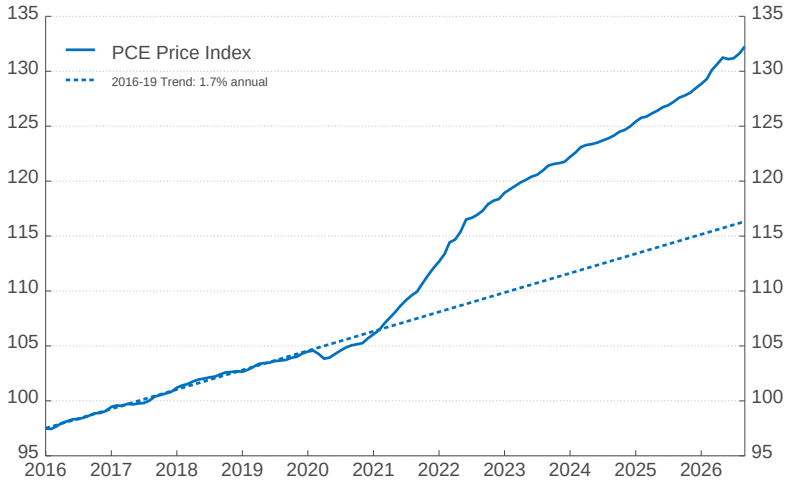
- Headline: 4.0%
- Ex Energy Goods: 3.1%
- Core: 3.3%

PCE Ex Energy Goods inflation trend for 2023-26: 2.8%

Prices and Inflation

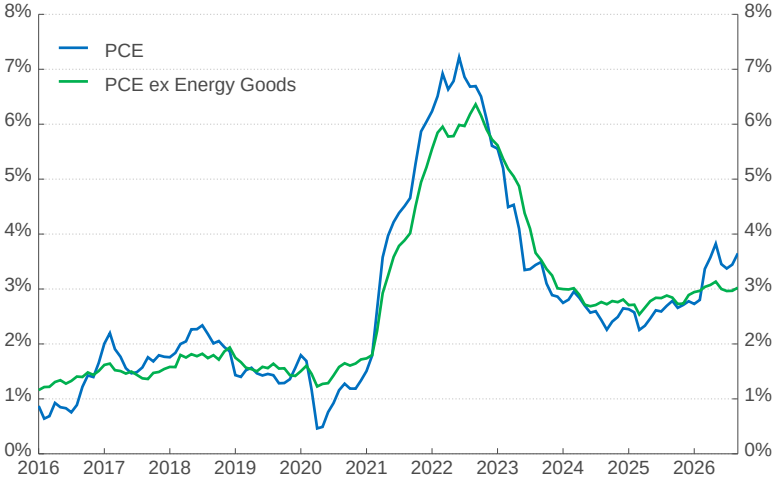
Price Level

Prices decline early in the pandemic and rise persistently above their trend afterwards



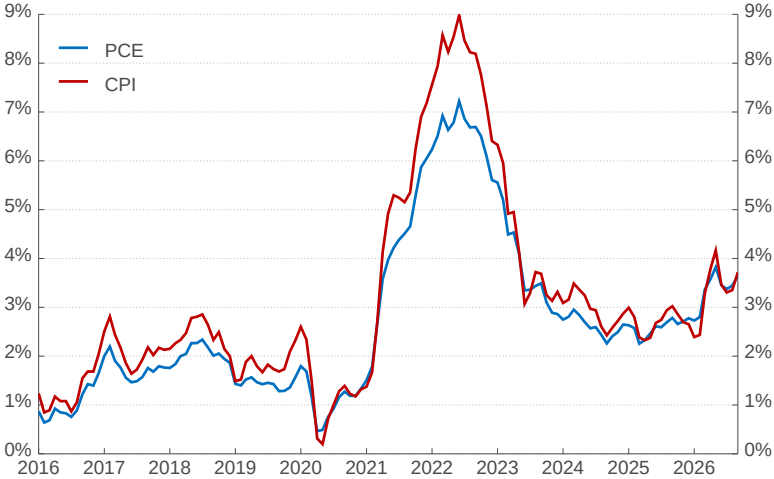
Inflation

Inflation has remained above 2% since March 2021—dynamics amplified by energy ▷ historical series



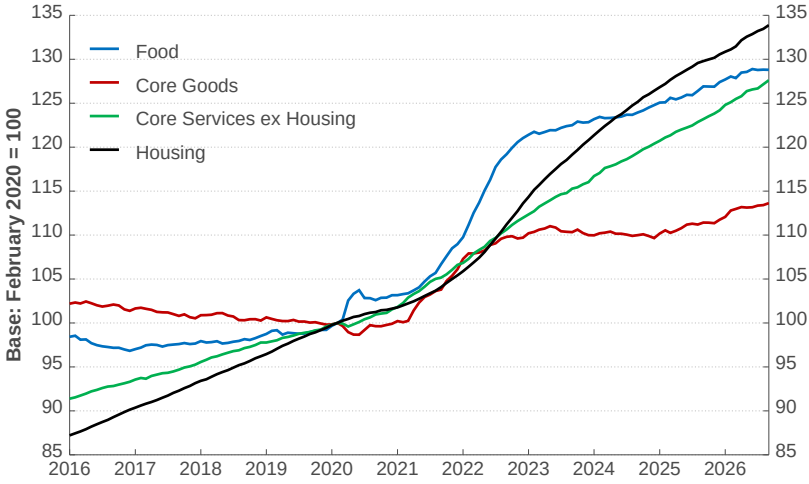
Comparing dynamics in PCE and CPI

▷ Excluding energy



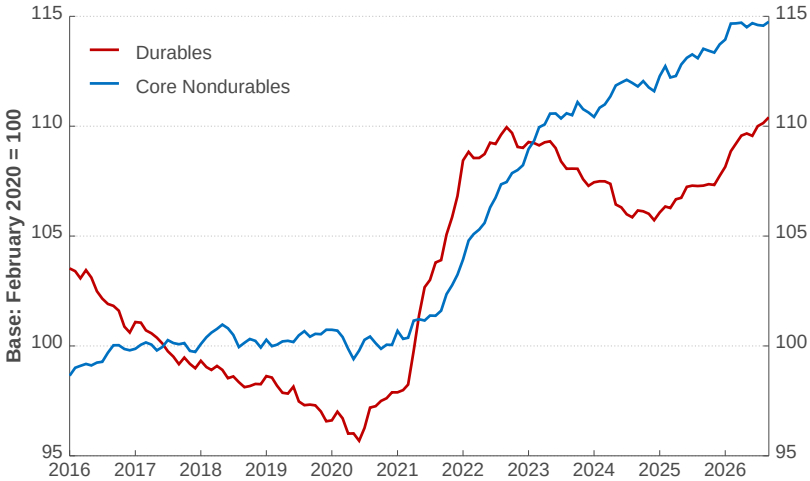
Components of PCE Price Index

Food, core goods, core services ex housing, and housing



Core Goods: Durables vs. Nondurables

Effect of tariffs evident on prices of core durable goods—core nondurables growing steadily



Inflation: aggregate measures

▹ CPI

Annualized PCE Inflation Rates

	All Goods & Services	Ex Energy Goods	Ex Food & Energy
2016–19	1.7%	1.6%	1.7%
2020	1.3%	1.7%	1.5%
2021	6.0%	5.2%	5.1%
2022	5.6%	5.7%	5.1%
2023	2.9%	3.0%	3.2%
2024	2.6%	2.8%	2.9%
2025	2.8%	2.9%	2.9%
2026 YTD	4.0%	3.1%	3.3%

SOURCES: Bureau of Economic Analysis and author's calculations.
NOTE: The 2026 period is the year through September.

Inflation: main components

▷ CPI

Annualized PCE Inflation Rates by Categories

	Food	Energy	Core Goods	Core Services ex Housing	Housing
2016–19	0.2%	4.2%	−0.6%	2.2%	3.4%
2020	4.0%	−7.5%	0.1%	2.0%	2.2%
2021	5.6%	30.8%	6.2%	5.0%	3.7%
2022	11.1%	6.6%	3.4%	5.0%	7.8%
2023	1.5%	−2.1%	0.3%	3.6%	6.3%
2024	1.6%	−1.1%	−0.3%	3.8%	4.7%
2025	2.1%	1.9%	1.9%	3.2%	3.3%
2026 YTD	1.5%	26.4%	2.3%	3.7%	3.4%

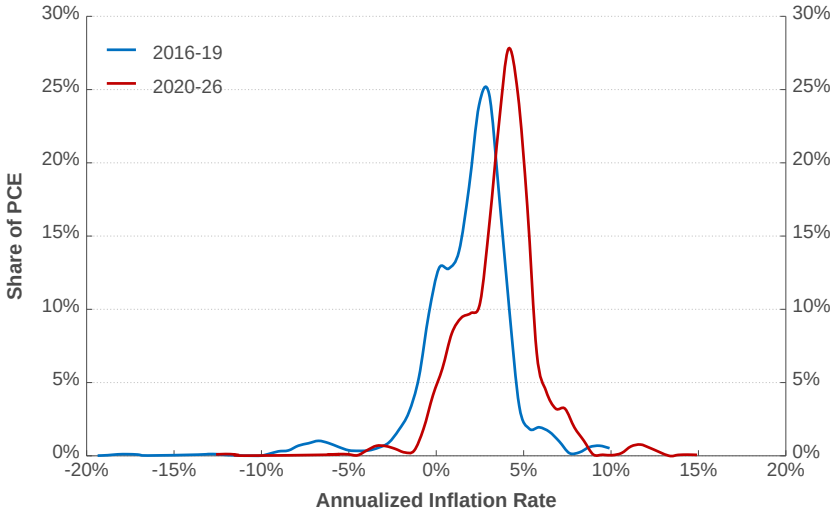
SOURCES: Bureau of Economic Analysis and author's calculations.
NOTE: The 2026 period is the year through September.

Inflation Distributions

Inflation increases have been widespread

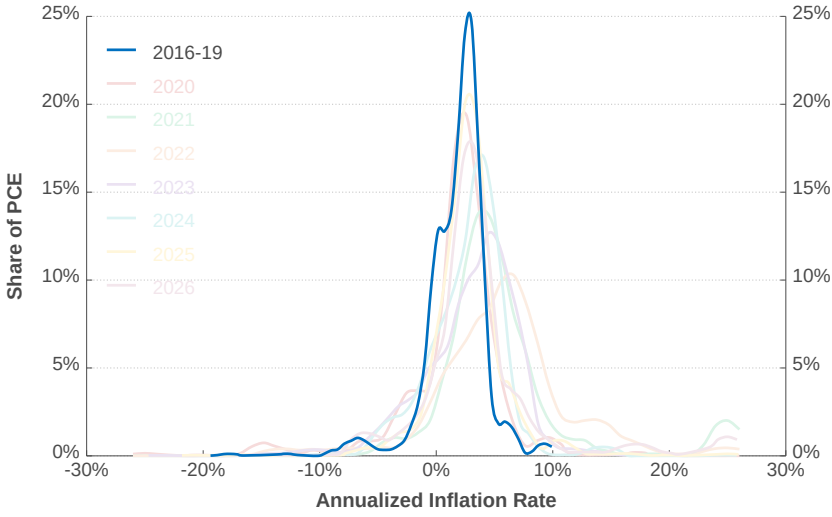
Methodology

Estimated distribution of PCE inflation ▷ core



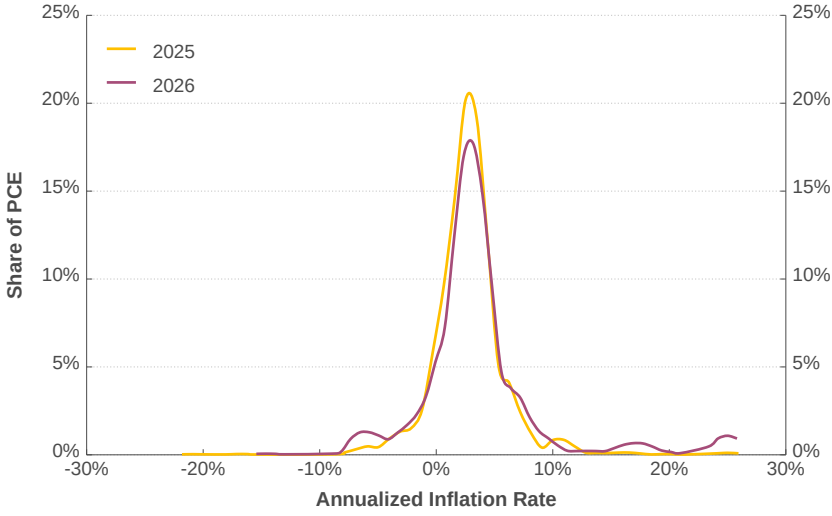
Inflation distributions: progression from 2016-19 to 2026

Estimated distribution of PCE inflation



Inflation Distributions: 2025 vs. 2026

Estimated distribution of PCE inflation ▷ core

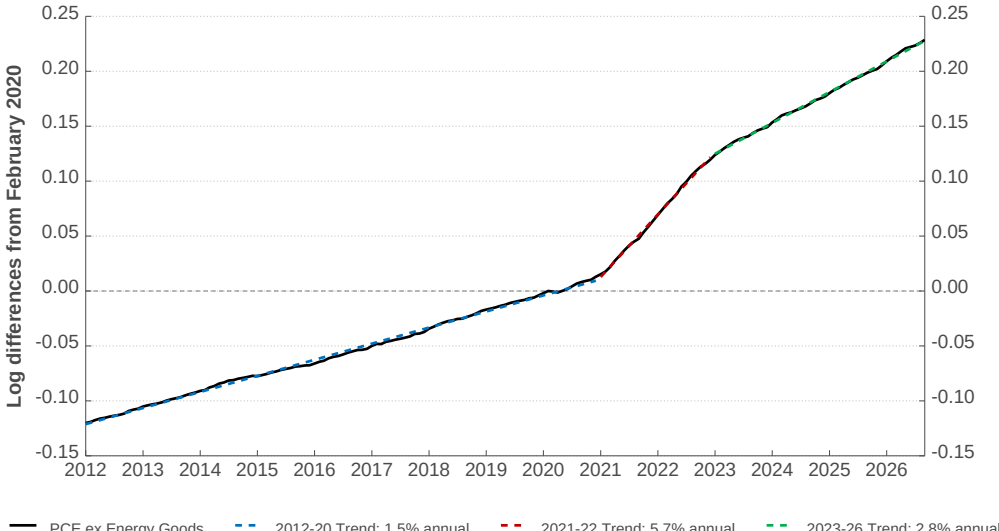


Inflation Regimes

Price Level Dynamics: Three regimes

Blog

(1) Below-Target Inflation: 2012-20 (2) Great Inflation: 2021-22 (3) Above-Target Inflation: 2023-26



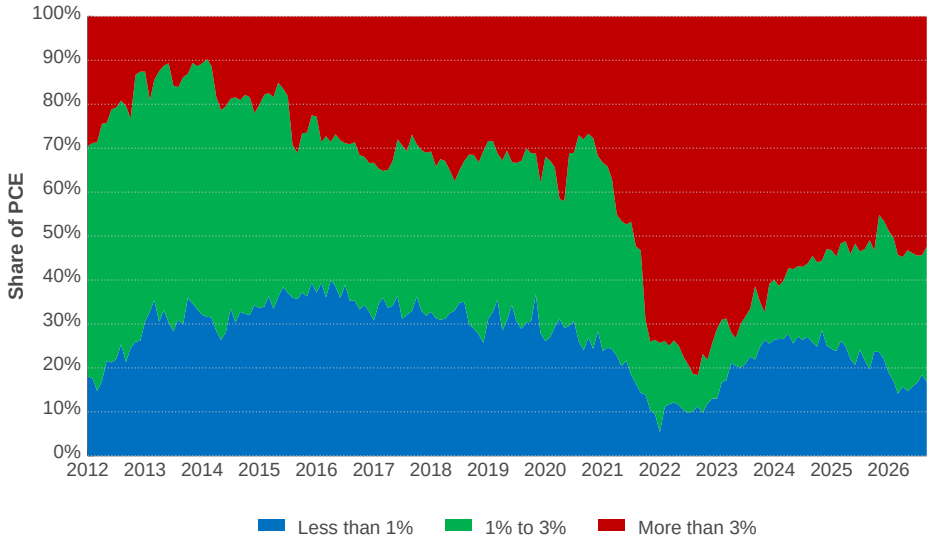
Dynamic PCE 12-Month Inflation

Products regrouped monthly by their current 12-month inflation ▷ ex energy goods



Dynamic PCE Expenditure Shares

Products regrouped monthly by their current 12-month inflation ▷ ex energy goods



Average PCE Expenditure Shares

Products regrouped monthly by their current 12-month inflation ▷ ex energy goods

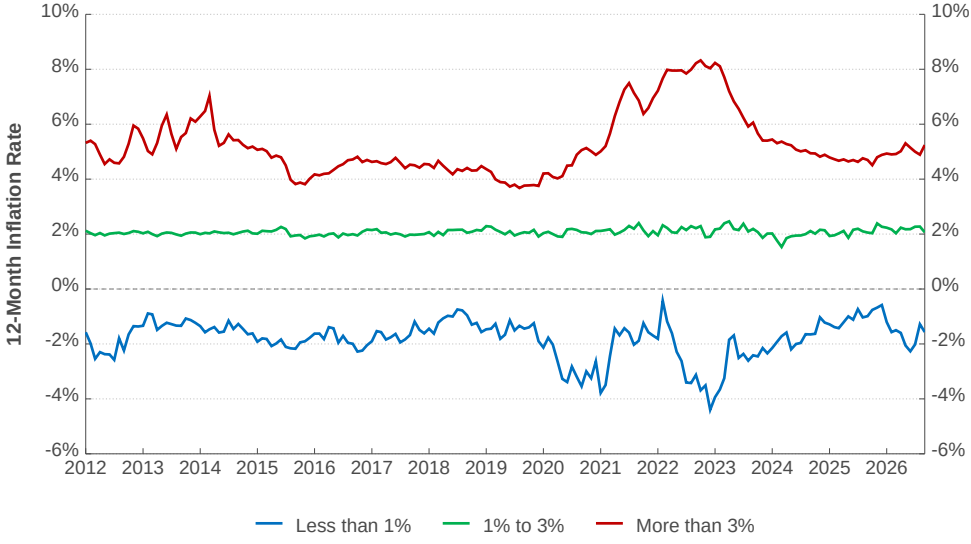
Average Shares of Nominal PCE

	Less than 1%	1% to 3%	More than 3%
2016–19	33.2%	35.6%	31.3%
2020	27.7%	40.1%	32.2%
2021	18.4%	30.6%	51.0%
2022	10.7%	12.6%	76.7%
2023	20.9%	11.4%	67.8%
2024	26.5%	16.4%	57.1%
2025	23.1%	25.3%	51.6%
2026	16.4%	30.6%	53.0%

NOTE: Arithmetic averages of monthly expenditure shares. 2026 covers January–September.

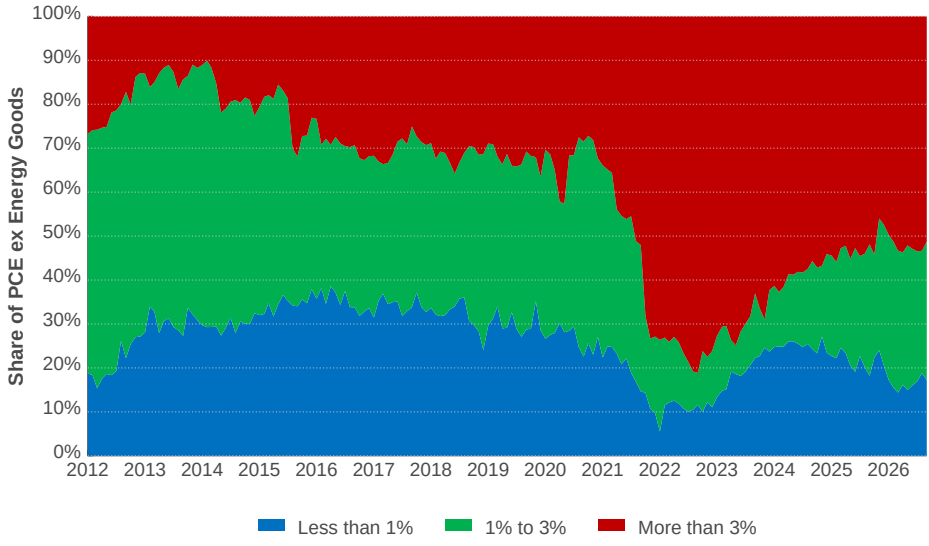
Dynamic PCE ex Energy Goods 12-Month Inflation

Products regrouped monthly by their current 12-month inflation ▷ total PCE



Dynamic PCE ex Energy Goods Expenditure Shares

Products regrouped monthly by their current 12-month inflation ▷ total PCE



Average PCE ex Energy Goods Expenditure Shares

Products regrouped monthly by their current 12-month inflation ▷ total PCE

Average Shares of Nominal PCE ex Energy Goods

	Less than 1%	1% to 3%	More than 3%
2016–19	32.9%	36.4%	30.8%
2020	26.8%	40.9%	32.4%
2021	18.6%	31.1%	50.2%
2022	10.8%	12.9%	76.2%
2023	19.3%	11.2%	69.4%
2024	25.0%	16.6%	58.4%
2025	21.7%	25.7%	52.6%
2026	16.4%	31.3%	52.4%

NOTE: Arithmetic averages of monthly expenditure shares. 2026 covers January–September.

Long-term inflation expectations

Markets vs. Models vs. Forecasters

10-Year Inflation Expectations — Period Averages

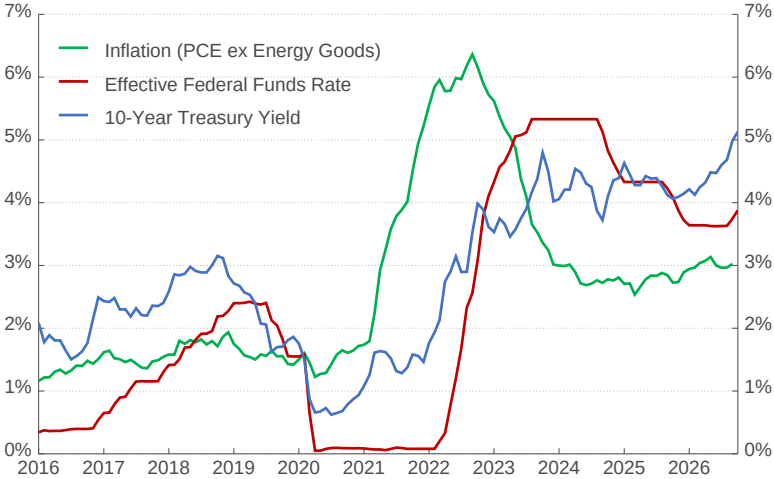
	TIPS	Swaps	Cleveland	DKW	SPF
2012–20	1.9%	2.2%	1.7%	2.2%	2.0%
2021–22	2.4%	2.7%	1.9%	2.2%	2.3%
2023–26	2.3%	2.5%	2.3%	2.7%	2.2%
Change 2012-20 to 2023-26	0.4%	0.3%	0.6%	0.5%	0.2%

SOURCES: FRED, Federal Reserve Board, Federal Reserve Bank of Philadelphia, Bloomberg and author's calculations.
NOTE: The 2026 period is the year through September.

Interest Rates

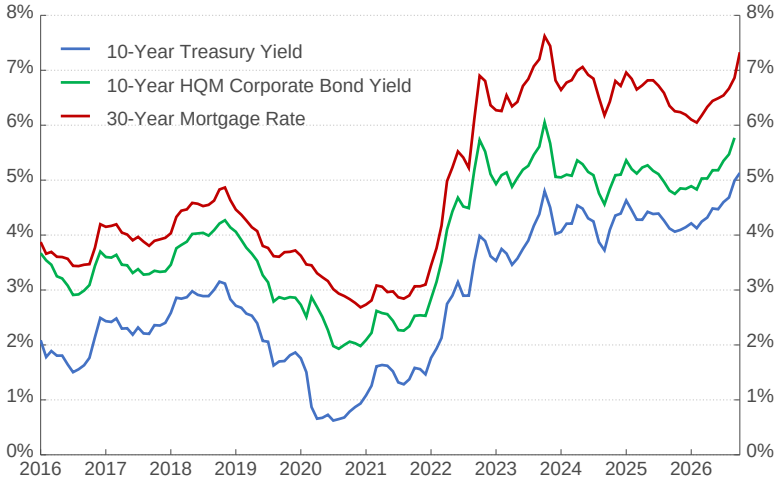
Monetary Policy

Easing cycles in September-December 2024 and September-December 2025



Monetary policy and long-term rates

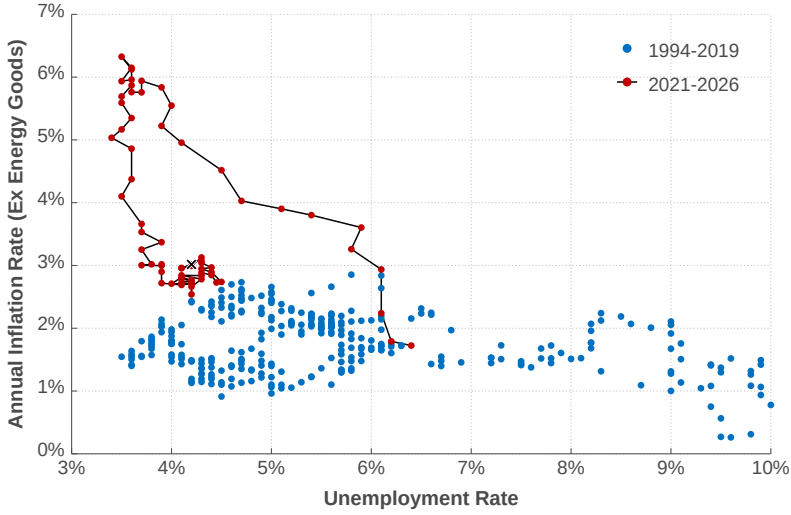
Annual rates



Phillips Curve

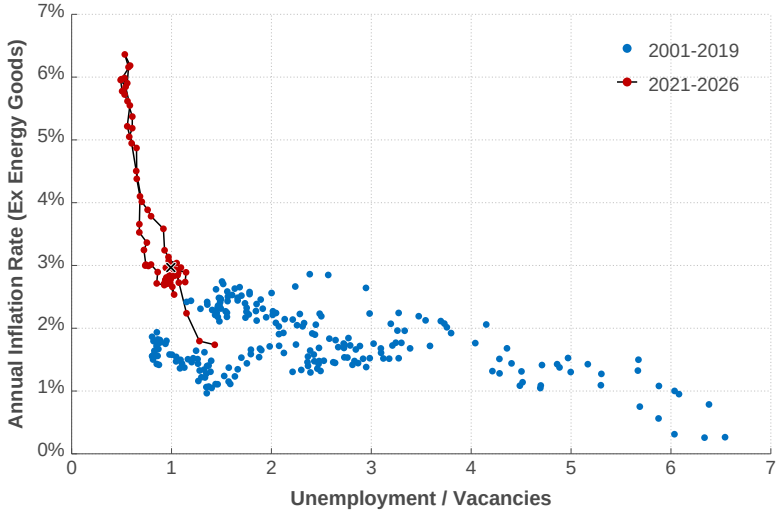
The Phillips Curve

Inflation vs. Unemployment



Alternative Phillips Curve

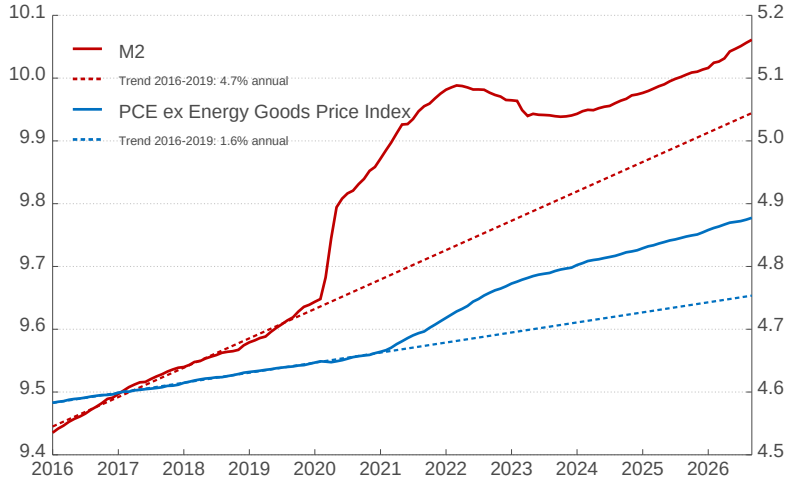
Inflation vs. Unemployment-to-Vacancies Ratio



$$\mathbf{M} \times \mathbf{V} = \mathbf{P} \times \mathbf{Y}$$

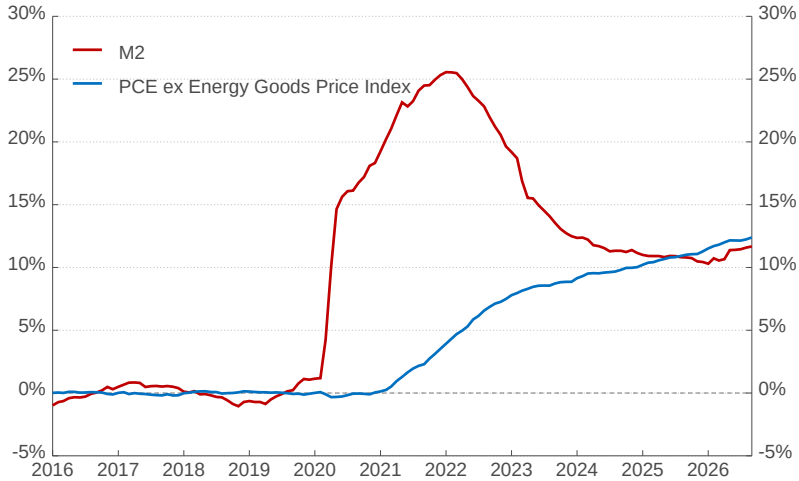
Money vs Prices: monetary expansion during COVID

In logs: M2 on left axis, prices on right axis—M2 grew at same rate as disposable income pre-COVID



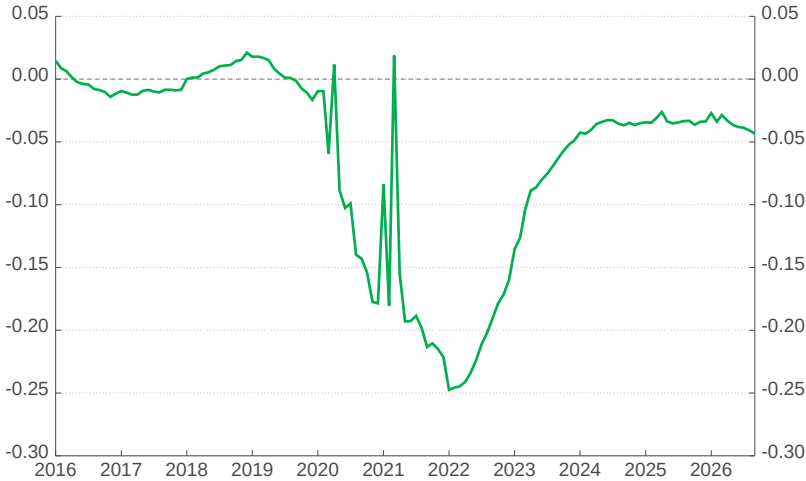
Price increase vs. monetary expansion

Deviations from pre-COVID trend



Velocity

Ln Velocity = Ln M2 - Ln Disposable Personal Income (Base: 2016-19 = 0)



Appendix

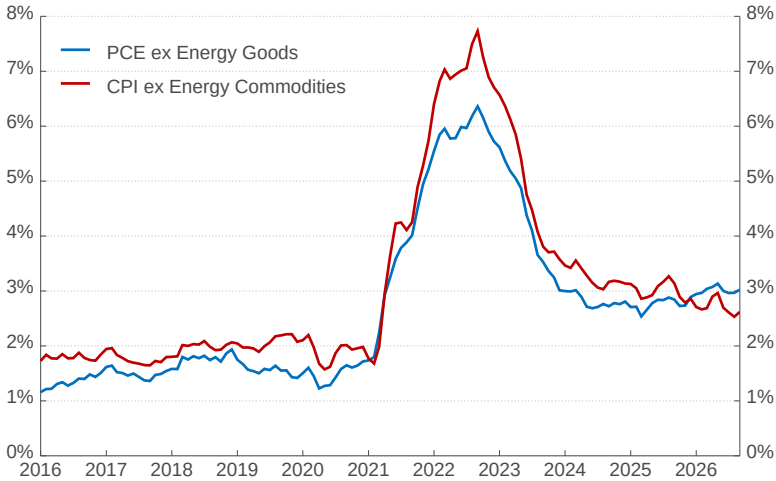
Current inflation in historical context

PCE ex energy goods annual inflation ▷ back



PCE ex energy goods and CPI ex Energy Commodities

▷ BACK



CPI Inflation Aggregates

▷ BACK

Annualized CPI Inflation Rates

	All Goods & Services	Ex Energy Commodities	Ex Food & Energy
2016–19	2.1%	1.9%	2.1%
2020	1.3%	2.0%	1.6%
2021	7.2%	5.7%	5.5%
2022	6.4%	6.7%	5.7%
2023	3.3%	3.6%	3.9%
2024	2.9%	3.1%	3.2%
2025	2.7%	2.9%	2.6%
2026 YTD	4.2%	3.5%	2.9%

SOURCES: U.S. Bureau of Labor Statistics and author's calculations.
NOTE: The 2026 period is the year through September.

CPI Inflation Components

▷ BACK

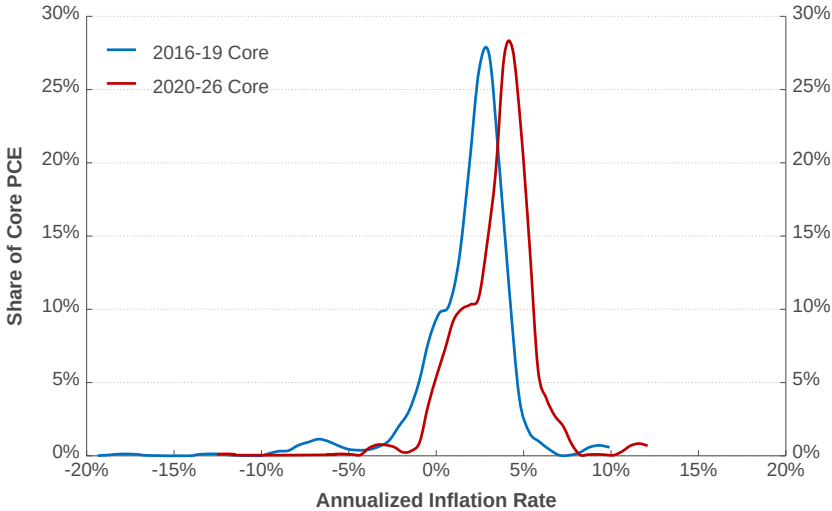
Annualized CPI Inflation Rates by Categories

	Food	Energy	Core Commodities	Services Less Rent	Rent of Shelter
2016–19	1.2%	3.9%	−0.2%	2.2%	3.3%
2020	3.9%	−7.0%	1.6%	1.4%	1.9%
2021	6.0%	30.3%	10.7%	3.9%	4.2%
2022	10.1%	6.9%	2.1%	7.5%	7.6%
2023	2.7%	−1.9%	0.1%	3.5%	6.2%
2024	2.4%	−0.3%	−0.7%	4.0%	4.6%
2025	3.0%	2.1%	1.4%	3.4%	3.1%
2026 YTD	2.2%	25.1%	0.8%	3.4%	3.5%

SOURCES: U.S. Bureau of Labor Statistics and author's calculations.
NOTE: The 2026 period is the year through September.

Core inflation increases have been widespread

Estimated distribution of Core PCE inflation ▷ back



Core Inflation Distributions: 2025 vs. 2026

Estimated distribution of Core PCE inflation ▷ back

